

Bridging the Gap: Translating AI Regulations into Socio-Technical Measures for Responsible AI

Tallal Ahmad

The Information School
University of Wisconsin-Madison
Madison, Wisconsin, USA
tahmad4@wisc.edu

Abstract

AI regulations and ethical frameworks are growing globally, but developers often struggle to translate them into practical action. My dissertation investigates how AI practitioners implement responsible AI principles in real-world settings, why current ethical toolkits fall short, and how governance practices can better reflect cultural and organizational realities especially in the Global South. Through three qualitative studies, I explore practitioner experiences, fairness in non-Western contexts, and co-develop actionable, context-aware AI governance frameworks. Feedback from the consortium will help refine this work to better align high-level policy with everyday development practices.

CCS Concepts

• **Social and professional topics** → *Governmental regulations.*

Keywords

Responsible AI; AI governance; socio-technical systems; AI ethics; fairness; Global South; AI policy; ethical frameworks; AI toolkits

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1 Introduction

The governance of artificial intelligence (AI) presents complex, multidimensional challenges that span technical, ethical, and regulatory domains. As AI technologies become increasingly embedded across various sectors, there is a heightened public and institutional demand for responsible and ethical deployment, emphasizing transparency, fairness, and accountability [3, 4]. AI governance encompasses a range of regulatory frameworks and ethical guidelines designed to align technological applications with organizational strategies, societal values, and legal standards [7, 8, 10, 20]. However, despite significant progress in establishing these frameworks, there remains a substantial gap between high-level regulatory principles

and their practical implementation in AI development processes [17, 18].

Current literature indicates that although numerous ethical frameworks and standards have been developed to guide Responsible AI (RAI), translating these guidelines into actionable measures remains challenging. Existing frameworks often articulate broad ethical commitments, such as transparency, fairness, and accountability—yet frequently lack the specific, operational guidance required by AI and machine learning (ML) practitioners for everyday workflows [13, 15, 16]. Practitioners face significant hurdles in operationalizing abstract ethical principles within real-world environments, where competing priorities and resource constraints further complicate ethical decision-making [17].

Moreover, the proliferation of ethical toolkits, though extensive, often suffers from issues related to usability, effectiveness, and alignment with practitioners' practical needs [11, 13]. This misalignment arises due to insufficient involvement of AI developers during toolkit creation, thereby limiting their practical adoption and effectiveness in real-world settings [11, 21]. The technical opacity of AI systems often referred to as the "black box" problem—further exacerbates these implementation challenges, posing barriers to accountability and transparency in algorithmic decision-making [5, 8, 9].

Adding complexity, fairness and ethics in AI are predominantly shaped by Western norms and legal standards, neglecting critical contextual differences across global sociocultural landscapes [12, 19]. This Western-centric approach to AI fairness not only risks imposing culturally misaligned standards but also potentially deepens existing inequalities in non-Western contexts [1, 6]. Thus, there is an urgent need to contextualize AI governance practices, ensuring their relevance and effectiveness across diverse geopolitical realities.

This proposal seeks to bridge these identified gaps by **translating** abstract AI regulatory policies into practical, actionable Responsible AI measures for AI and ML developers and practitioners. By integrating regulatory insights with practitioner-oriented strategies, this research aims to facilitate more effective, inclusive, and culturally responsive AI governance, ultimately supporting the development of ethically robust AI systems aligned with societal values. The overarching research question guiding this dissertation is:

- How can regulatory AI governance policies and ethical principles be effectively translated into practical and actionable Responsible AI measures for AI and ML developers, considering diverse global contexts, technical complexities, organizational constraints, and socio-cultural influences?

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2 Ongoing Research

2.1 Study 1: Exploring Practitioners Experiences

This study examines how AI developers and practitioners engage with ethical frameworks and toolkits in real-world industry settings, particularly as these tools are increasingly positioned as mechanisms for achieving regulatory compliance. While ethical AI principles and toolkits have proliferated in response to societal concerns and policy shifts, many professionals struggle to apply them effectively. Challenges include poor usability, lack of regulatory alignment, and difficulties integrating ethical considerations into fast-paced development workflows.

By conducting in-depth interviews with 21 AI practitioners, this study explores how ethical frameworks are interpreted, adapted, or bypassed in practice—especially under growing legal and institutional pressure to demonstrate compliance with responsible AI standards. It aims to uncover the limitations and workarounds practitioners employ, the internal organizational mechanisms that support or hinder compliance, and what additional resources are needed to make these tools both ethically meaningful and legally actionable.

2.1.1 Preliminary Findings.

- **Prioritizing Legal Compliance:** Practitioners predominantly translate ethical guidelines into practice through direct compliance with legal and regulatory mandates, such as GDPR for data protection or FDA regulations in healthcare. Actions driven by these legal requirements often involve explicit instructions provided through organizational documentation, compliance checklists, or direct communication from dedicated legal teams, such as removing sensitive data from datasets.
- **Organizational Structures and Responsibility:** The responsibility for ethical compliance differs significantly based on organizational structure. In larger companies, specialized compliance teams, legal departments, or dedicated "red teams" typically oversee regulatory adherence, while developers usually participate through periodic training sessions (e.g., GDPR compliance). Day-to-day ethical oversight is primarily managed by these specialized units. In contrast, smaller startups generally assign regulatory compliance directly to developers themselves. Regardless of size, senior management frequently holds ultimate accountability for ensuring compliance. Across all organizations, building and maintaining internal expertise is emphasized as crucial for implementing sustainable ethical AI practices.
- **Challenges in Translation:** Practitioners frequently face significant challenges when attempting to translate ethical frameworks into actionable practices. One common issue is the gap between abstract guidelines and the practical realities of implementation, which can leave developers uncertain about how to operationalize ethical principles. Additionally, the lack of standardization and inconsistent metrics across different frameworks and toolkits complicates integration efforts. The fast-paced nature of industry work, coupled with limited time and resources, often hampers the thorough incorporation of ethical practices. Developers also report

tensions between ethical imperatives and business demands, including budget constraints and tight sprint cycles. Further barriers include insufficient documentation, a lack of clear examples or benchmarking data for toolkit usage, and the rapid evolution of the AI landscape, which makes it difficult for teams to stay current. Finally, the inherently subjective nature of ethics means that developers must frequently navigate complex, context-specific decisions without clear-cut answers, requiring ongoing consultation and guidance.

Despite the growing presence of ethical frameworks and regulatory policies aimed at guiding responsible AI development, preliminary findings reveal a significant gap in their effective translation into practice. This translation is often ambiguous, shaped by a range of organizational, technical, and contextual factors—including the availability and usability of toolkits, clarity of internal roles, regulatory interpretation, and the degree of training and documentation provided. Rather than a linear or standardized process, implementation tends to vary widely across settings, leading to inconsistencies and uncertainty. These findings highlight the need for more actionable and context-aware guidance, improved standardization of tools and metrics, and better alignment between high-level regulatory goals and the day-to-day realities of AI development.

3 Proposed Studies

3.1 Study 2: Examining AI Fairness and Ethical AI Governance in the Global South

Artificial Intelligence (AI) is increasingly deployed across critical sectors in the Global South. These sectors include healthcare, education, and criminal justice. AI is shaping decision-making processes with profound social and ethical implications. While AI holds the potential to enhance efficiency, accessibility, and service delivery, significant concerns remain. These concerns center around algorithmic fairness, bias, and ethical governance. Existing fairness frameworks and AI governance models have been largely developed in Western contexts. These models may not adequately capture the complex realities of the Global South. The sociotechnical, economic, and historical complexities of these regions are often overlooked. This study seeks to address these gaps. It aims to explore how fairness is conceptualized, implemented, and experienced across different stakeholder groups in the Global South. Additionally, the study will examine the governance challenges surrounding AI deployment in resource-constrained, socio-politically diverse environments.

This study will employ a qualitative, multi-stakeholder approach to investigate AI fairness and ethical governance in the Global South, utilizing virtual semi-structured interviews to gather in-depth perspectives from approximately 20–30 participants across multiple countries. Recruitment will target a diverse range of key stakeholders, including regulators/policymakers who shape AI frameworks, technologists (developers, engineers, data scientists) involved in AI system design and deployment, and academics and advocacy groups providing critical insights on ethics, fairness, and the experiences of marginalized communities; participants will be sought through professional networks, snowball sampling, institutional outreach, and freelance platforms. The interview guides will be structured around the study's theoretical framework, covering

philosophical roots, axes of discrimination, and legal framing, to ensure comprehensive exploration of how fairness is understood, implemented, and experienced within the socio-political and cultural contexts of resource-constrained environments in the Global South.

3.2 Study 3: Co-Developing Actionable AI Governance Frameworks

Building on the findings from the first two studies, the third study will take a participatory approach by organizing hands-on workshops that bring together policymakers, regulators, and AI practitioners. These workshops will serve as a space for collaborative problem-solving, where participants will engage in structured discussions, scenario analyses, and policy co-design exercises. The goal is to co-develop actionable AI governance frameworks that bridge the gap between technical and socio-cultural perspectives on AI fairness and ethics. By integrating insights from AI practitioners on real-world implementation challenges and policymakers' perspectives on regulatory feasibility, this study aims to create policy recommendations that are inclusive, context-sensitive, and practically implementable.

This study adopts the Social Construction of Technology (SCOT) framework [14] to collaboratively develop AI governance frameworks, emphasizing stakeholder interactions and interpretive flexibility in shaping technology. Utilizing participatory research methods [2], the study will engage diverse stakeholders—including policymakers, AI practitioners, regulators, advocacy groups, and users—in a series of structured workshops. These workshops will systematically explore, document, and negotiate varied stakeholder meanings, expectations, and technological imaginaries regarding AI governance. Data collection will be comprehensive, using audio-recorded sessions, field notes, policy drafts, scenario exercises, and post-workshop interviews, alongside participant reflections and institutional documents. Analysis will involve constructivist thematic methods guided by SCOT, employing open, axial, and selective coding to uncover relationships and power dynamics in how different groups interpret and negotiate AI governance. The outcome aims to produce a nuanced understanding of the social construction and potential stabilization of AI governance frameworks through stakeholder collaboration.

4 Expected Contribution

This dissertation will contribute to the fields of AI governance, responsible AI development, and socio-technical systems. The expected outcomes focus on translating high-level AI regulations and ethical frameworks into practical tools and processes that support developers in implementing responsible AI in real-world settings. Study 1 will generate a practitioner-informed framework that improves the usability and regulatory alignment of ethical AI toolkits. Study 2 will expand current understandings of algorithmic fairness by centering perspectives from the Global South, addressing the cultural and institutional gaps in Western-centric models. Study 3 will co-develop actionable, context-sensitive AI governance strategies through stakeholder collaboration, producing practical guidelines that reflect both policy and development realities. Collectively, this work aims to make AI systems more transparent, accountable, and

culturally responsive, offering practical pathways for bridging the gap between policy and practice in global AI development. Insights from this research will inform future regulatory designs, toolkit development, and interdisciplinary collaborations for ethical and inclusive AI governance.

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